

Name: _____

Date: _____

RATIO, (SIMPLIFYING RATIOS)

LO: I can simplify ratios by finding common factors, including ratios with decimals and fractions.

Simplifying ratios

A ratio is in its **simplest form** when both sides share no common factor other than 1. Divide both sides by the **highest common factor (HCF)**.

Tip: For decimals, multiply to make whole numbers first. For fractions, find a common denominator or multiply by LCM.

Quick examples

●	12:8 = 3:2	divide both by HCF 4
●	0.5:1.5 = 1:3	multiply both by 2
●	$\frac{1}{2}:\frac{3}{4} = 2:3$	multiply by 4
●	12:8 = 6:3	6:4 then 3:2
●	0.5:1 = 5:1	should be 1:2

Key idea

Divide both sides by the HCF. For decimals or fractions, make whole numbers first.

$$12:18 \div \text{HCF}(6) = 2:3$$

Always express in simplest integer form

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - integers

Simplify 24:36.

HCF of 24 and 36 = 12

$$24 \div 12 = 2$$

$$36 \div 12 = 3$$

2:3

Divide both numbers by their HCF.

Example 2 - decimals

Simplify 0.6:1.5.

Multiply both by 10: 6:15

HCF of 6 and 15 = 3

$$6 \div 3 = 2, 15 \div 3 = 5$$

2:5

Remove decimals first, then simplify.

Example 3 - fractions

Simplify $\frac{1}{3}:\frac{1}{2}$.

Common denominator = 6

Multiply both by 6: 2:3

Already in simplest form

2:3

Multiply both fractions by the LCM of denominators.

Example 4 - units

Write 20 minutes : 1 hour as a simple ratio.

Convert to same units: 20 min : 60 min

HCF of 20 and 60 = 20

$$20 \div 20 = 1, 60 \div 20 = 3$$

1:3

Always convert to the same units before simplifying.

YOUR TURN – PRACTICE (deliberate practice)

1. Simplifying integer ratios

Simplify each ratio.

- a) 6:10
- b) 15:25
- c) 18:24
- d) 42:56

2. Decimal ratios

Simplify each decimal ratio.

- a) 0.4:1.2
- b) 1.5:2.5
- c) 0.8:0.6
- d) 2.4:3.6

3. Fraction ratios

Simplify each ratio involving fractions.

- a) $\frac{1}{4}:\frac{3}{4}$
- b) $\frac{2}{3}:\frac{1}{6}$
- c) $\frac{3}{5}:\frac{2}{5}$
- d) $\frac{1}{2}:\frac{2}{3}:\frac{1}{4}$

4. Converting units

Convert to same units then simplify.

- a) 30 cm : 1 m
- b) 500 ml : 2 litres
- c) 45 min : 2 hours
- d) £1.50 : £4.50

5. Mixed simplification

Simplify each ratio fully.

- a) 48:64
- b) 0.75:1.25
- c) $\frac{2}{3}:\frac{5}{6}$
- d) 250 g : 1.5 kg
- e) 3.6:2.4:1.2

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To simplify a ratio, divide both sides by the HCF ☐

Correct answer: _____
Reason: _____

b) 0.5:1 simplifies to 5:1 ☐

Correct answer: _____
Reason: _____

c) Ratios must be in the same units before simplifying ☐

Correct answer: _____
Reason: _____

d) 12:8 simplifies to 6:3 ☐

Correct answer: _____
Reason: _____

e) $\frac{1}{2}:\frac{1}{3} = 3:2$ ☐

Correct answer: _____
Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A recipe uses **750 ml of milk**, **0.5 litres of cream**, and **250 ml of water**. Write the ratio of milk:cream:water in its simplest form.

Expression: _____

Simplified: _____

b) A map has scale **2 cm : 5 km**. Express this as a ratio in the form 1:n. Two towns are **8 km apart** in real life. What is the distance on the map?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

- Have I shown all my working clearly?
- Did I check my answer using a different method?
- Does my answer look reasonable?

